

	Work Instruction	Doc. Revision	1 of 1
	STD Machine Adjustable Rail Remove and Replace Procedure (Optical PIP)	Effective Date	03/29/2016

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
1	03/29/2016	First Release	Chin Chee Yan

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Remove Procedure

1. Unload all board from machine, go to panel handling, change the display unit as “Inch” and initialize, then manually set the rail width to **5 inch**.

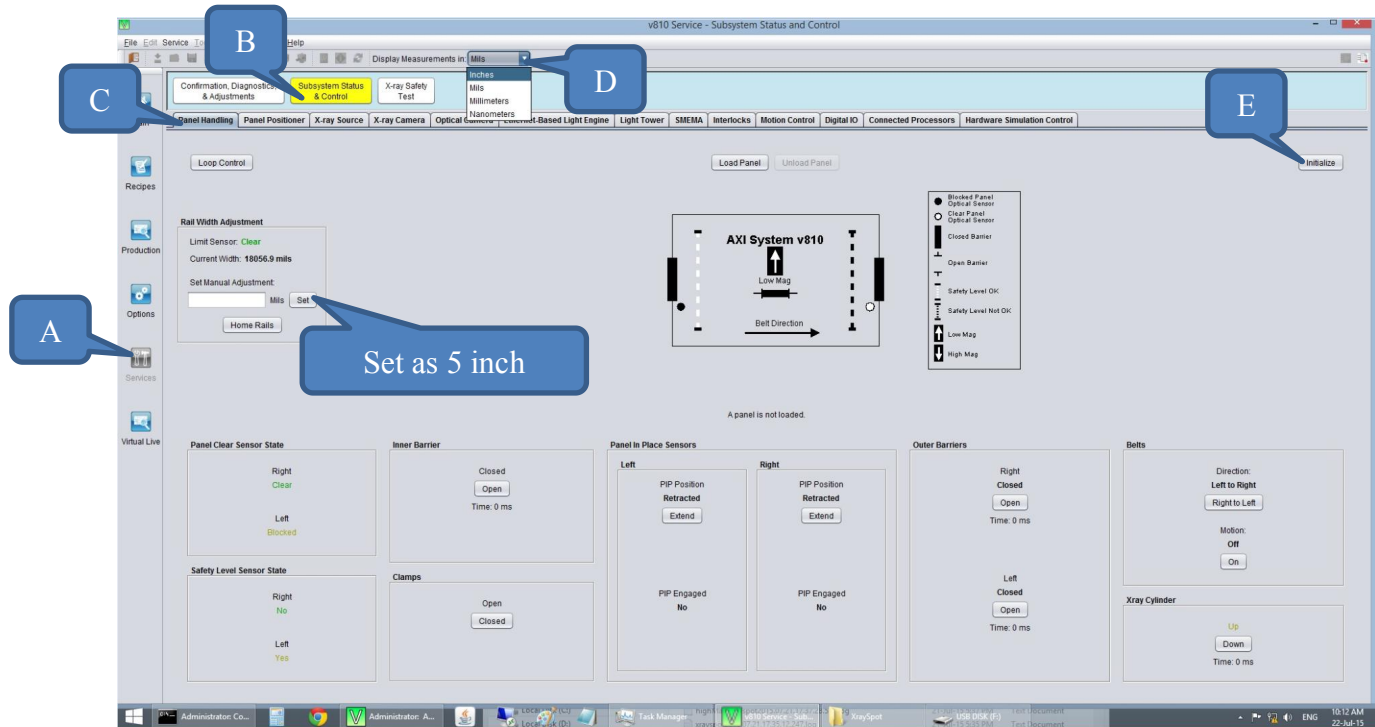


Figure 1

2. Turn off X-Ray and power down machine as usual. Turn off the main power and air inlet. Ensure SSCA power is turn off.
3. Open front, rear and right access door for easier task perform. Stage AWA width will similar as shown.



Figure 2

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4. At the Left Outer barrier, disconnect cables as shown, unscrew the temper proof screws and remove the Left Outer Barrier. Cautions, door assembly is extreme heavy, need 2 people to remove.

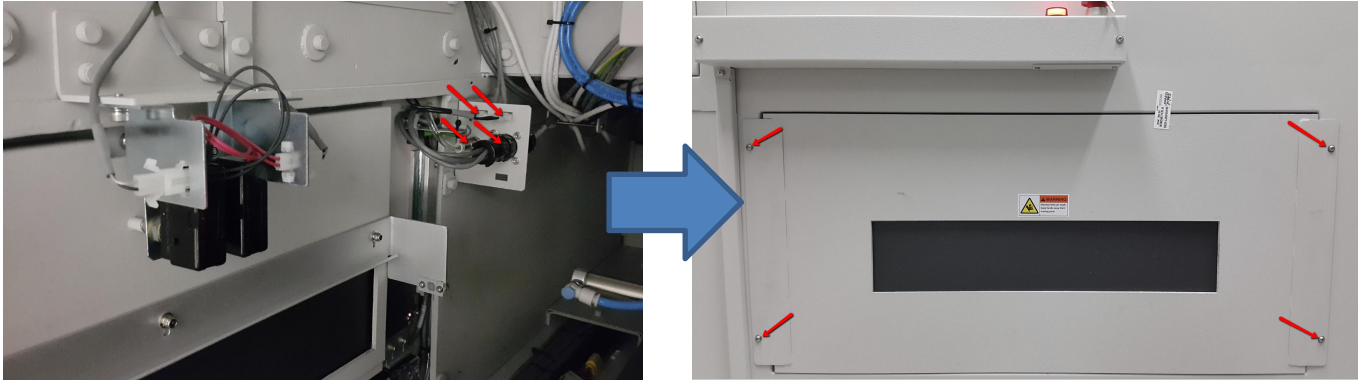


Figure 3

5. At the Right Outer barrier, disconnect cables as shown, unscrew the temper proof screws and remove the Right Outer Barrier. Cautions, door assembly is extreme heavy, need 2 people to remove.

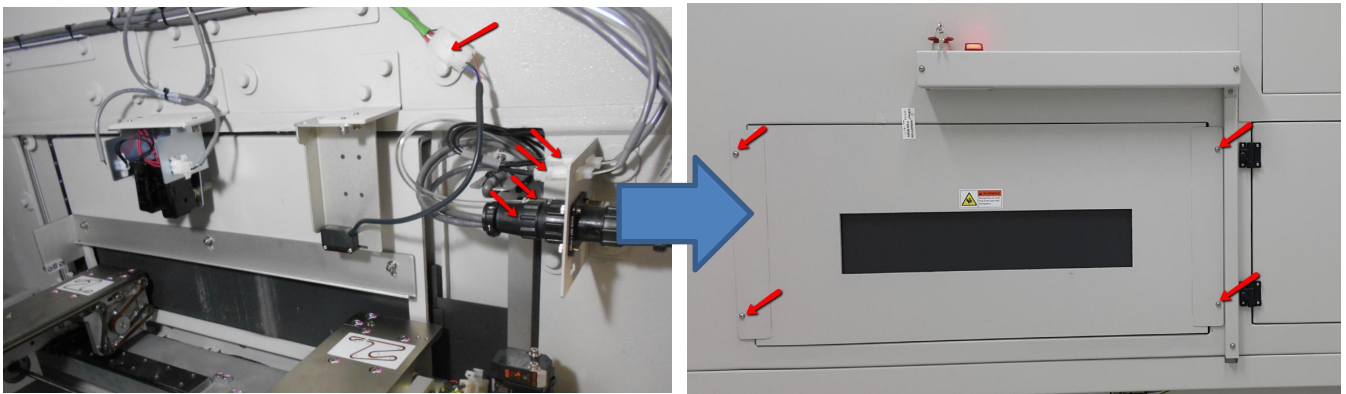


Figure 4

6. Disconnect the connect for DC motor kit (P/N, 2670-0033) at Adjustable rail as shown.

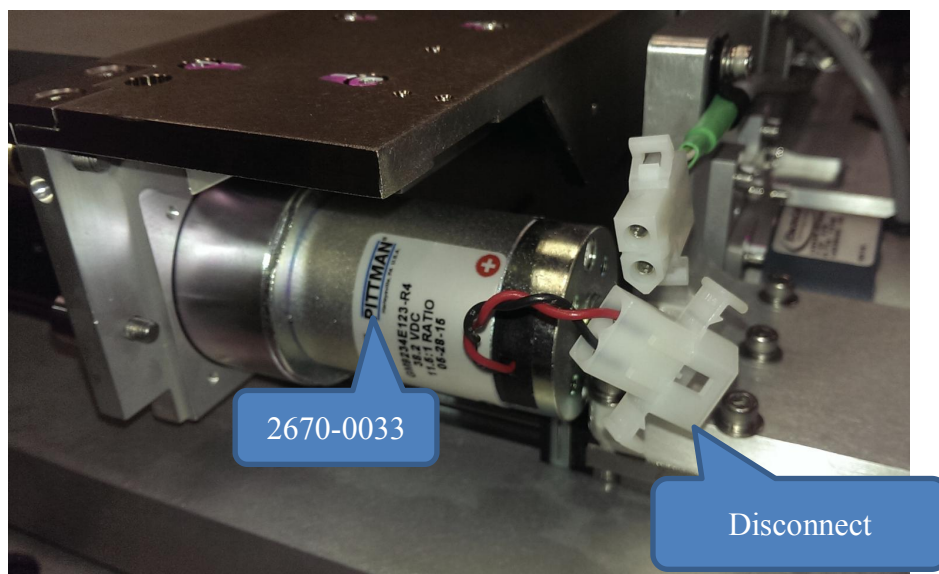


Figure 5

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7. Unscrew CABLE CLAMP, SCREW MOUNT, 3/16 (P/N, 2560-0001), **unlock** the Bearing_Oiless 1/8ID X .252OD (P/N, **2650-0108**) and **disconnect** the **both tubing** from FITTING, HOSE W/GASKET, 1/8 HOSE 10-32 (P/N, 2820-0005). Remark the orientation before remove

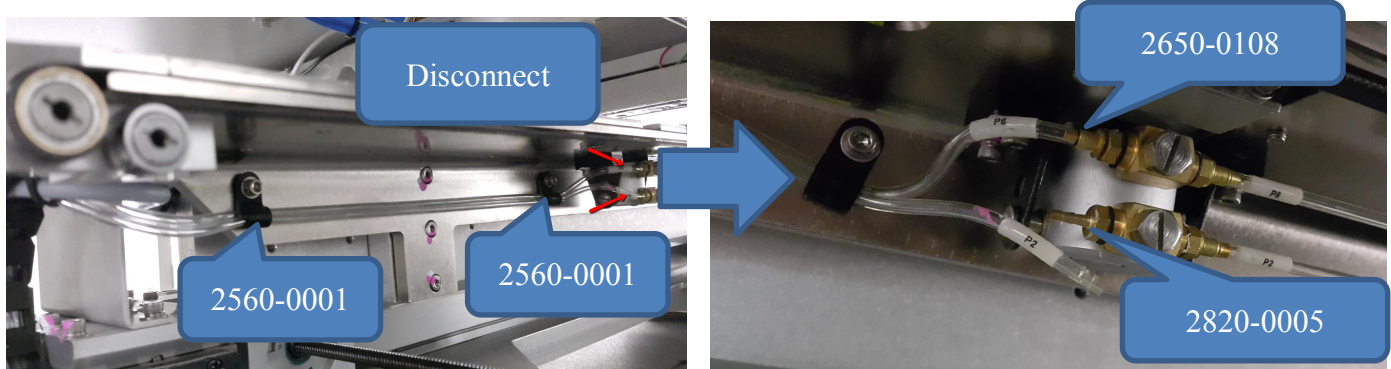


Figure 6

8. Remove cable tie and unscrew CABLE CLAMP, SCREW MOUNT, 3/16 (P/N, 2560-0001) and remove 2284-0051 from routing. Hanging in secure area so that we could reused after replacement.



Figure 7

9. Break the V810 Auto Width E-Chain (P/N, 2000-0001) very 1st joint as shown, place those cable nicely.

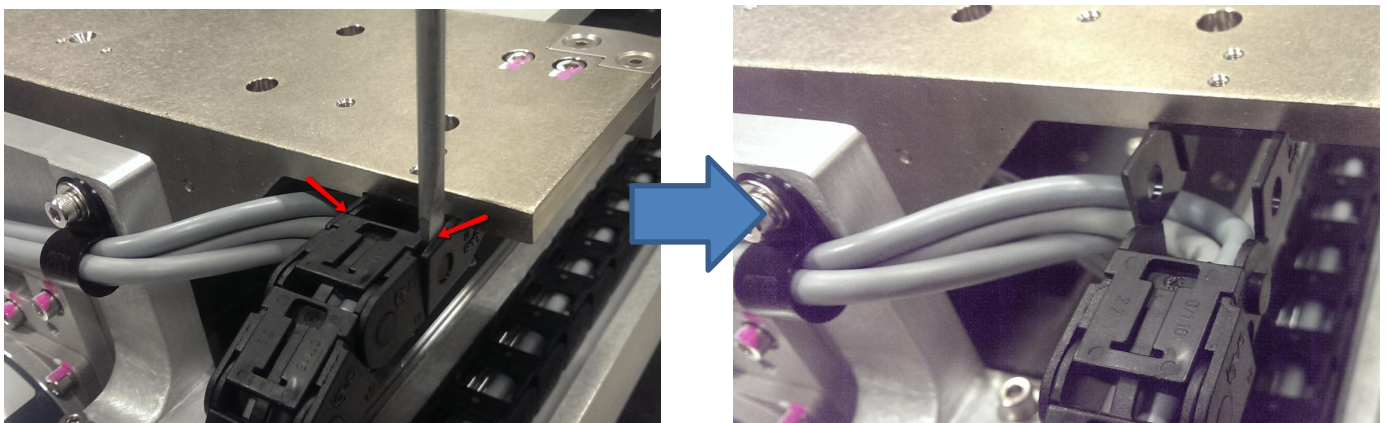


Figure 8

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10. Unscrew (SHCS M4x16 (8x)) as shown and complete remove the old fix rail kit from stage.

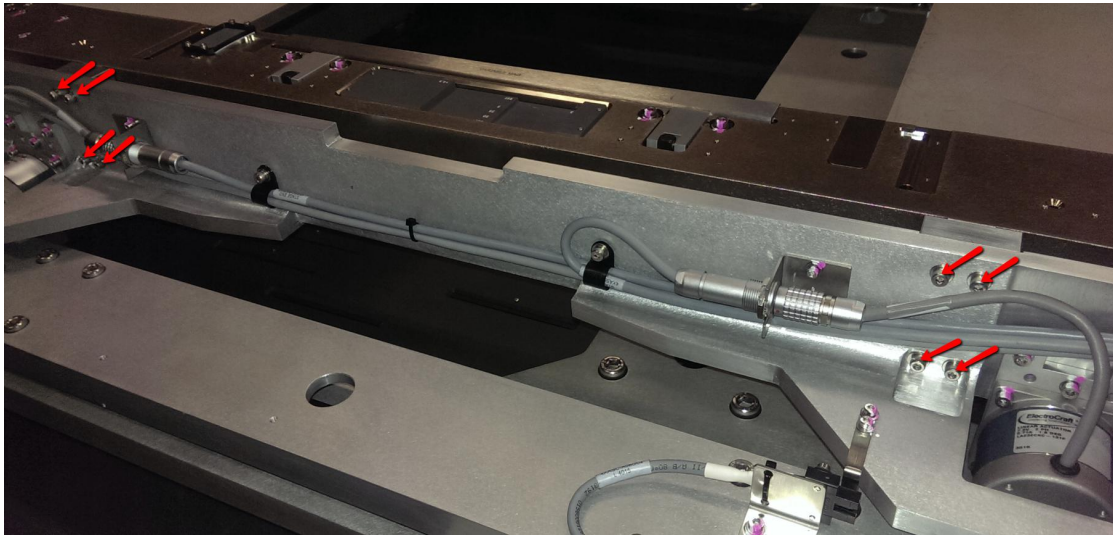


Figure 9

11. Unscrew as shown and remove the Mount, Conveyor Block x 2 (P/N, 4XAXY-009) to new Fix Rail Assembly.

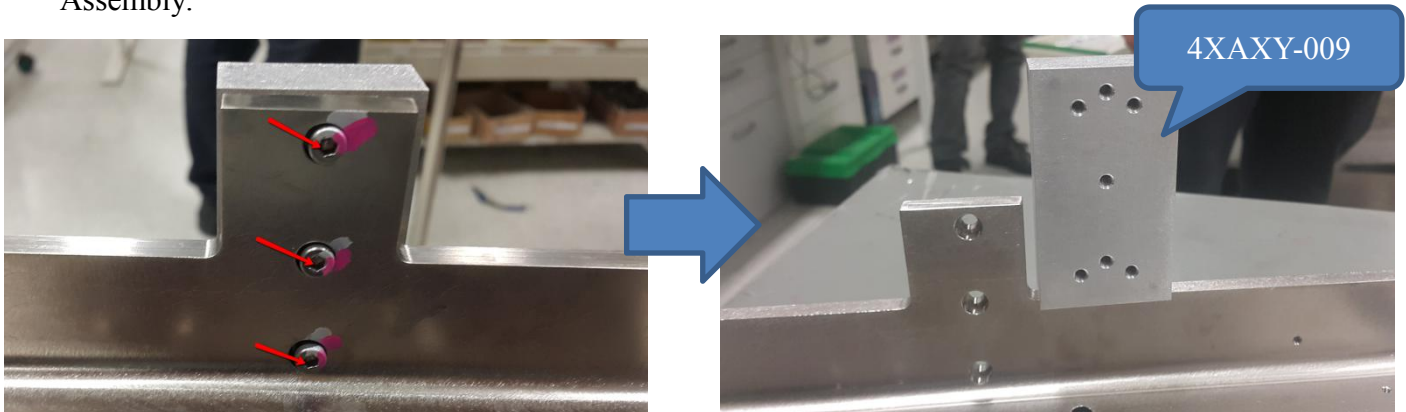


Figure 10

12. Unscrew and remove Shield, Board Drop (P/N, 4XAGR-031) for front side as shown.



Figure 11

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Rail Height Alignment Jig Setup

13. Prepare the Rail Height Alignment Jig (P/N, 4XAHW-A107) as shown and set-up.

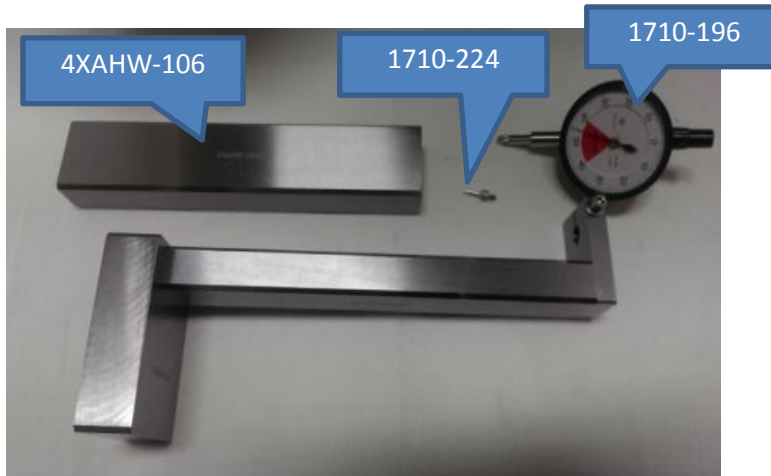


Figure 12

14. Assemble the Lever Points (P/N, 1710-224) to Mitutoyo Dial Indicator (P/N, 1710-196) as shown.

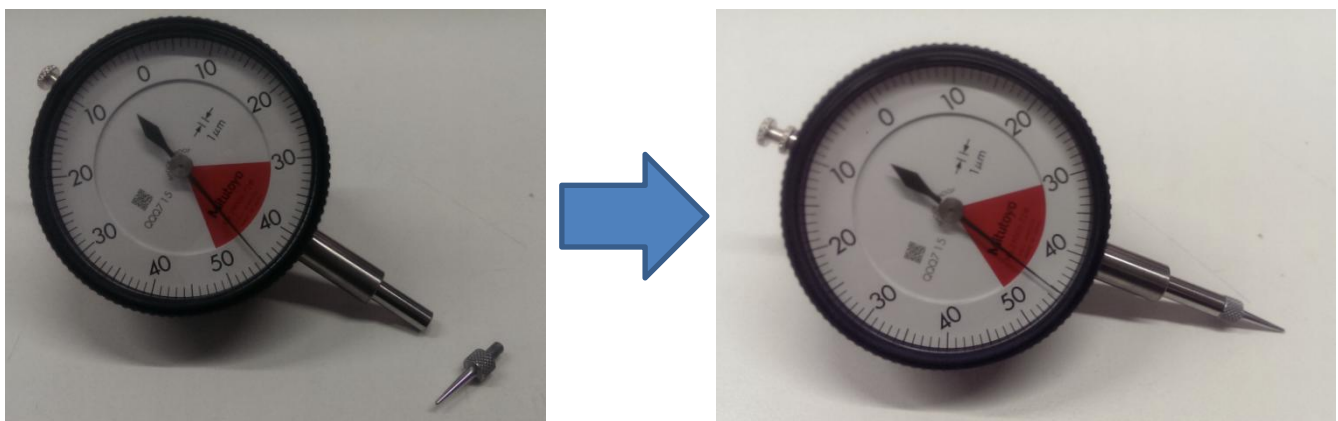


Figure 13

15. Ensure all screws at jig are tighten.

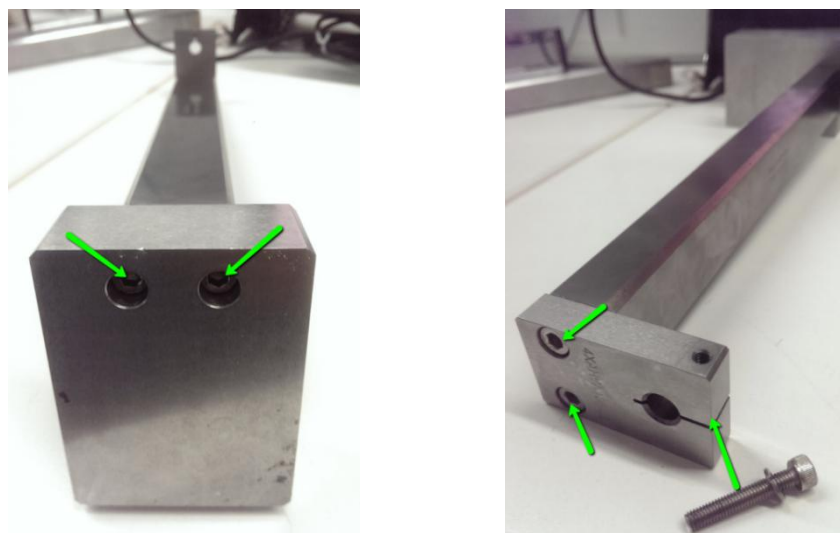


Figure 14

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16. Place the Mitutoyo Dial Indicator to jig as shown, caution on screw insert direction.

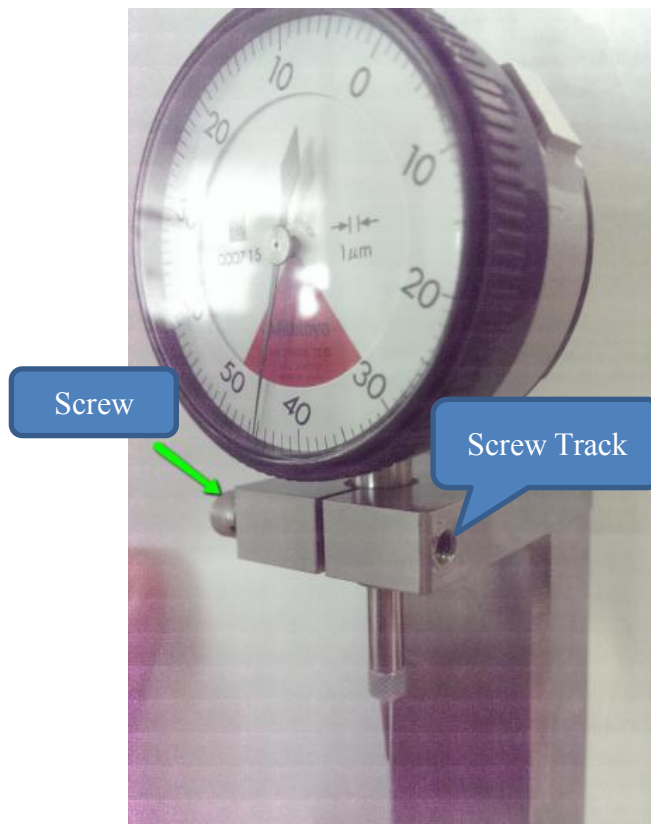


Figure 15

17. Place the jig at machine granite and set-up the jig (Mitutoyo Dial Indicator) height as shown. Adjust the Dial Indicator until the needle point touching center of Rail Height Calibration Block Gage (P/N, 4XAHW-106) surface. Ensure when the the gauge pull or remove the 4XAHW-106, indicator still can play around. **Once done, tighten the screw and no adjustment any more for accuracy.**

Gauge can only play 1 round (0.1mm), so when adjust the gauge height, try to get the needle shown at center point of gauge.

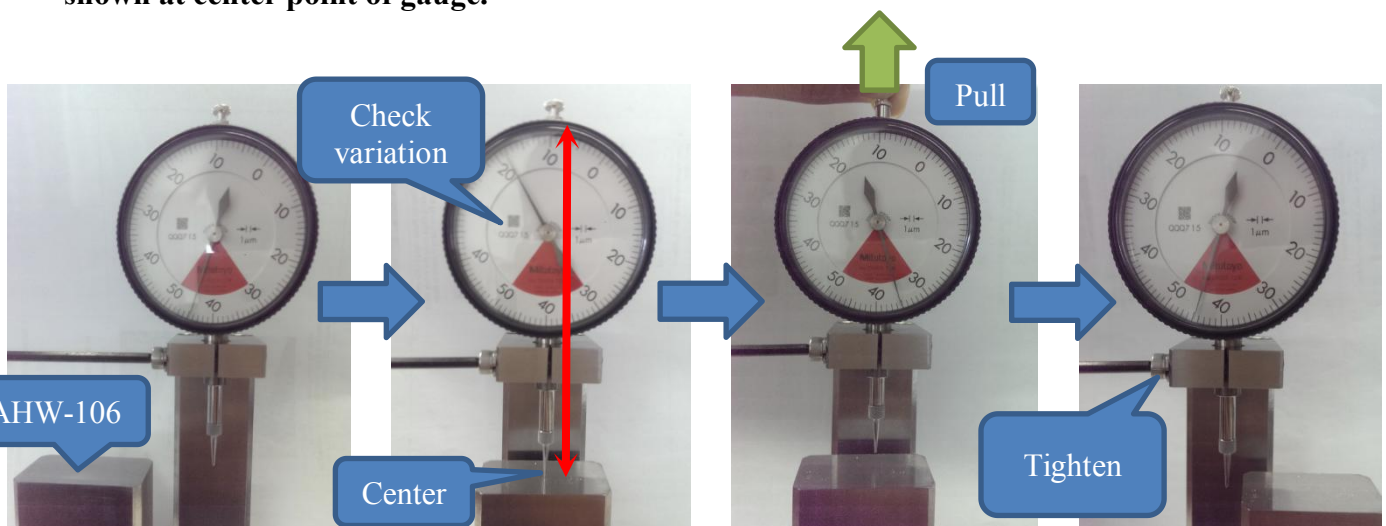


Figure 16

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18. Once done, place 4XAHW-106 and touching the dial indicator and rotate the gauge meter to “0” for easier visual. **Gentle remind, the value set will be the reference to calibrate rail height.**

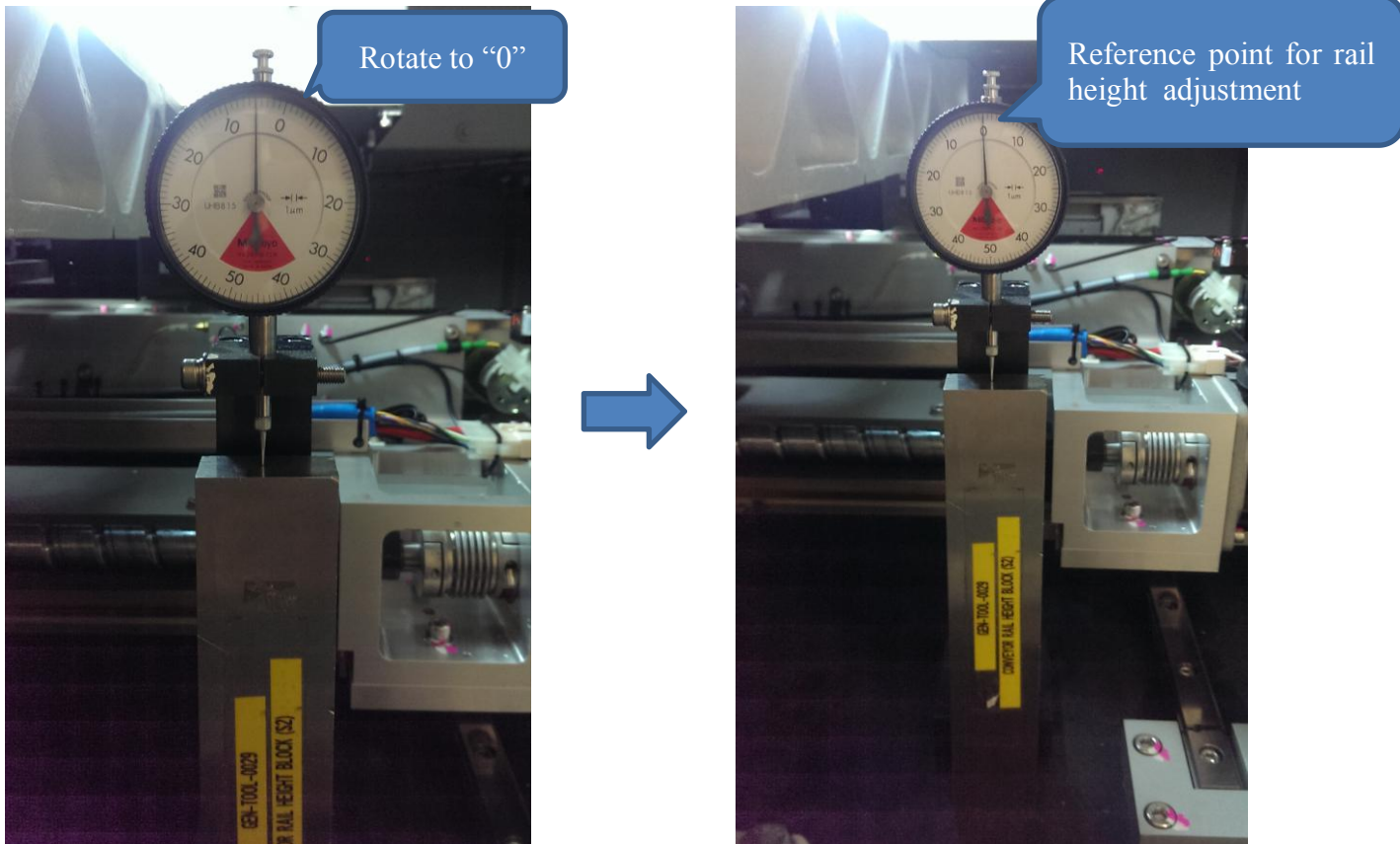


Figure 17

Replace Procedure

19. Prepare a new ASSEMBLY, CONVEYOR RAIL, ADJUSTABLE (P/N, 4XAXY-093).

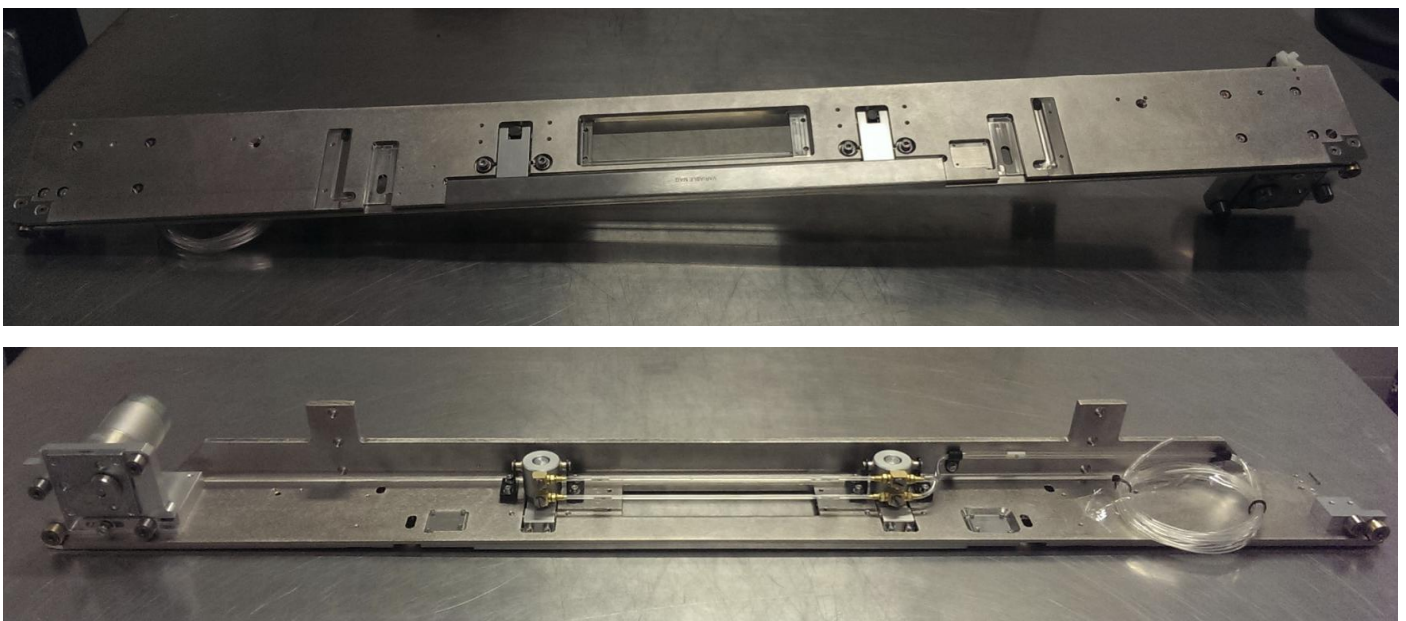


Figure 18

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20. Remove the assembled tubing as shown.

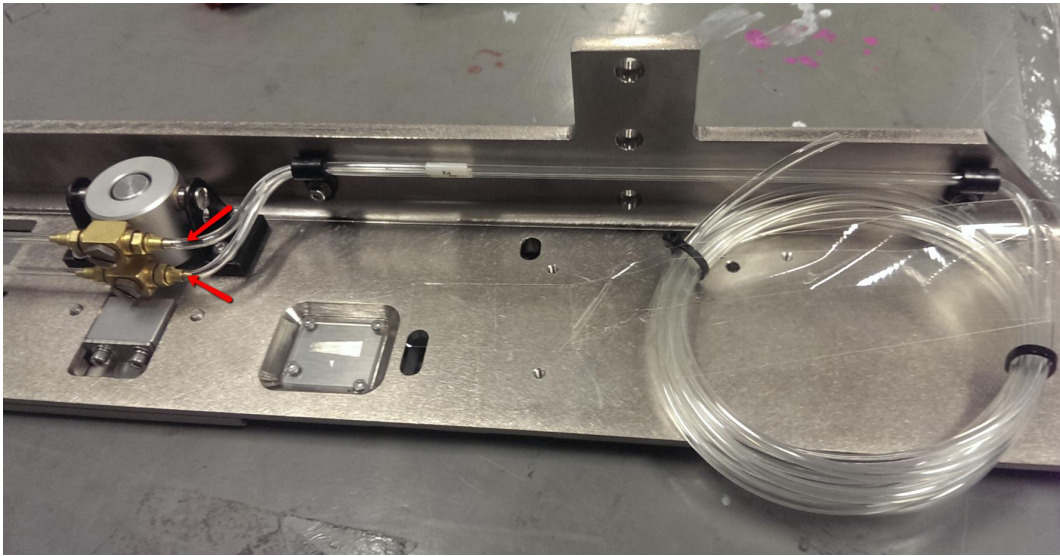


Figure 19

21. Transfer the E-chain end bracket from old Adjustable rail kit to new 4XAXY-093.

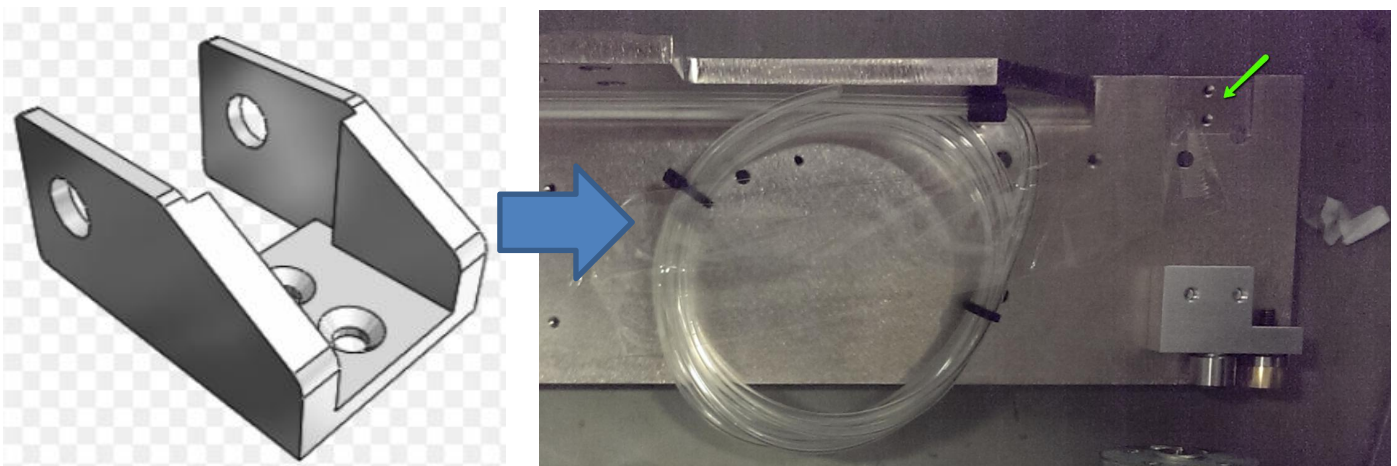


Figure 20

22. Transfer the 4XAXY-009 x 2 to new 4XAXY-093, caution the orientation when assembly.

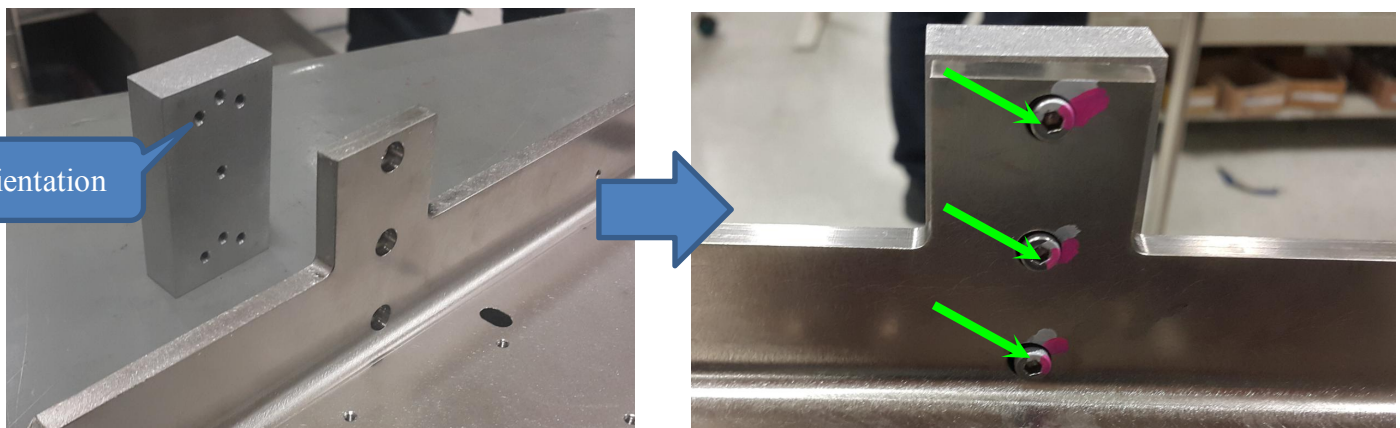


Figure 21

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23. Use L-Ruler to align the 4XAXY-009 with 4XAXY-093 top surface as shown, ensure the contact surface height gap is smaller as possible. Once done align, tighten the screws.

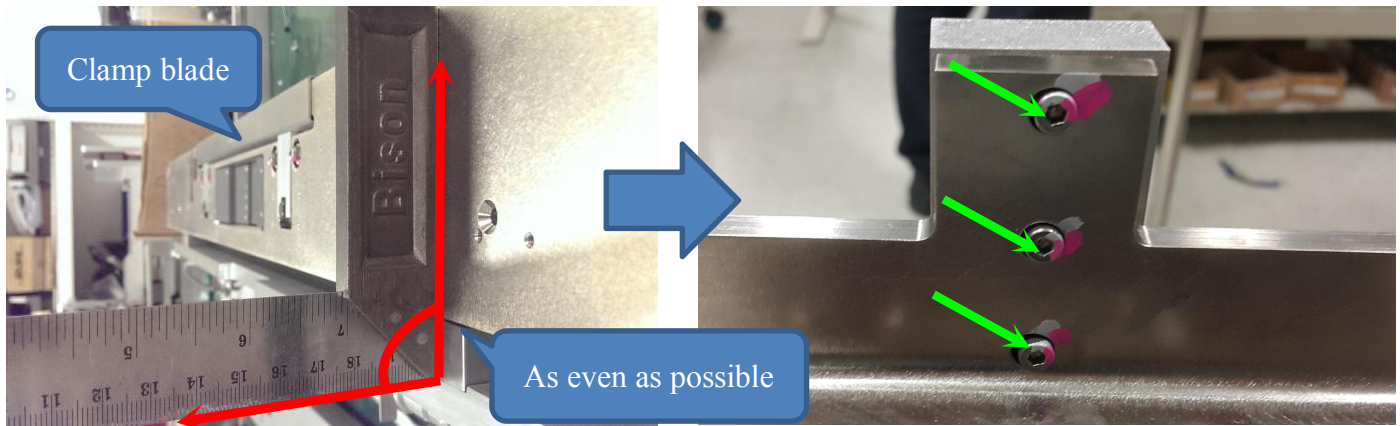


Figure 22

24. Assembly the new 4XAXY-093 to stage. Place the Rail Height Calibration Block Gage (P/N, 4XAHW-106) at granite near to Right side. **Adjust the 4XAXY-093 height approximately with same height with 4XAHW-106 as shown, then hand tight the screws. Perform same step for left side of 4XAXY-009. Both site screws MUST be hand tight with less variation. Caution, ensure the cylinder nose is well align when assembly.**

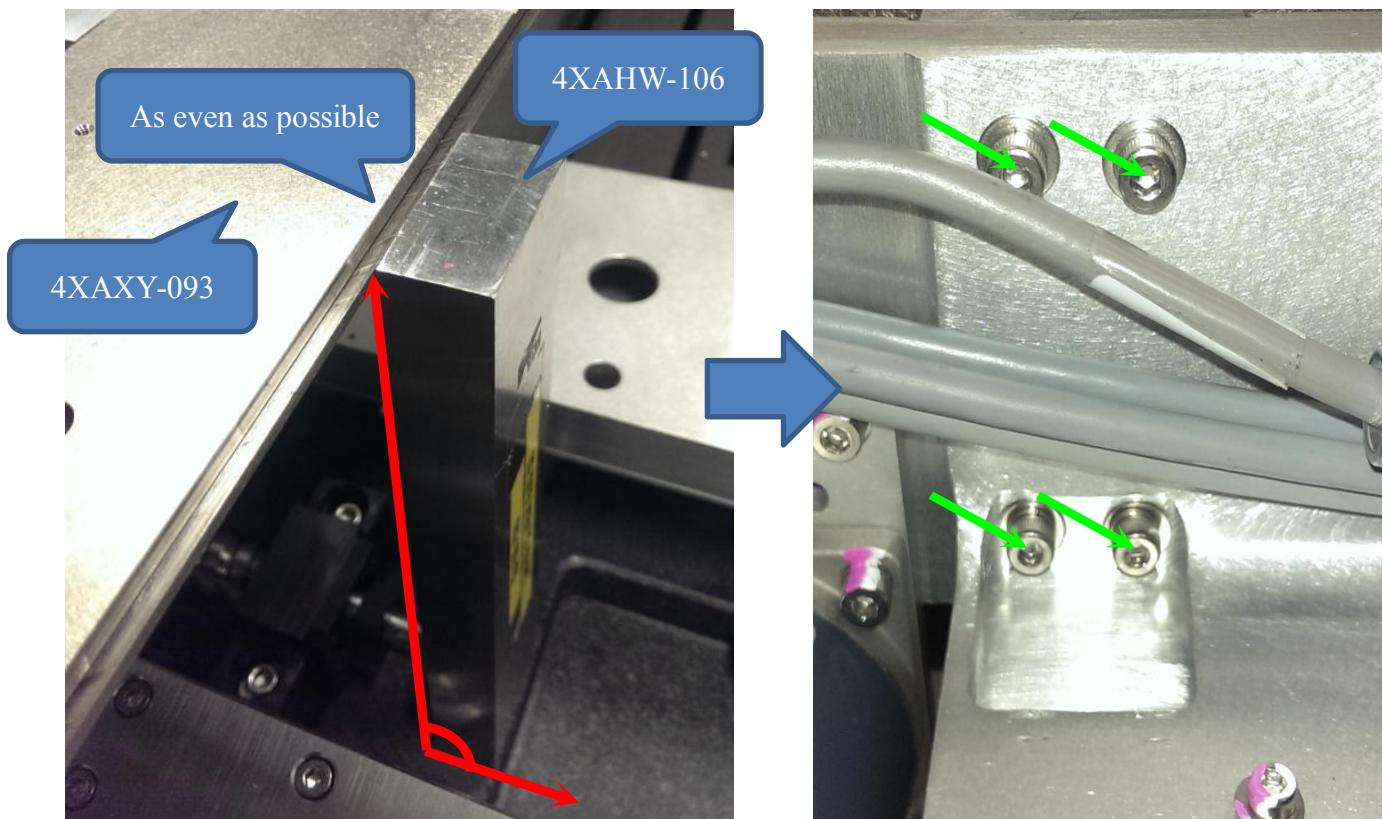


Figure 23

25. Clean the stage and from debris.

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26. Start with Right side, place the 4XAHW-A107 as shown, ensure the jig base is even to the granite.

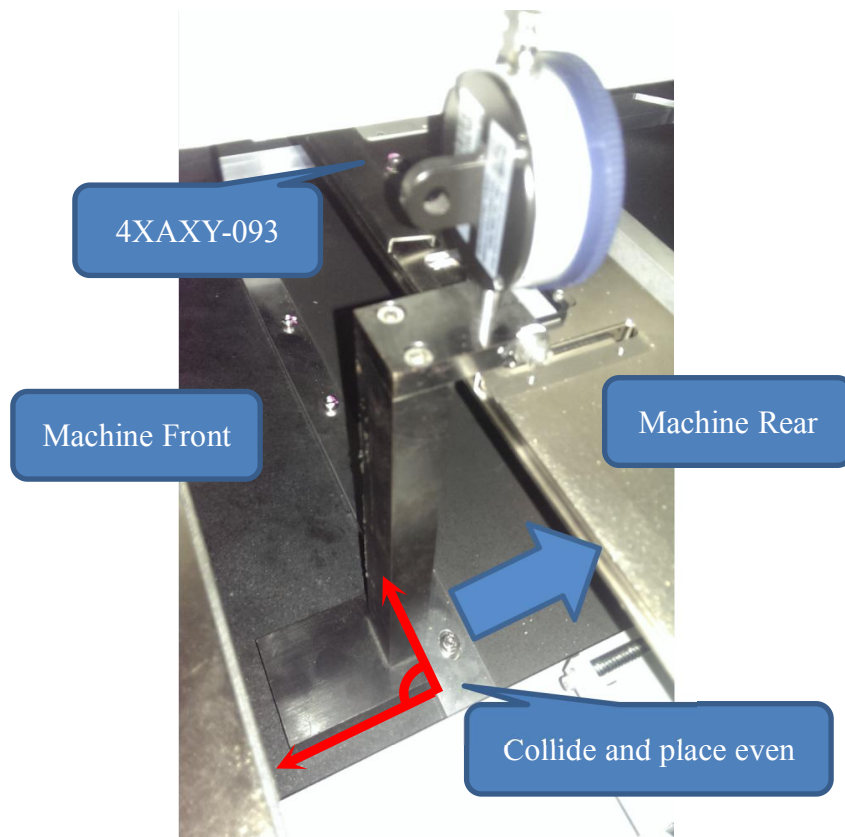


Figure 24

27. Manually move the stage until the needle point probe at inner rail surface in front on Right PIP opening as shown.

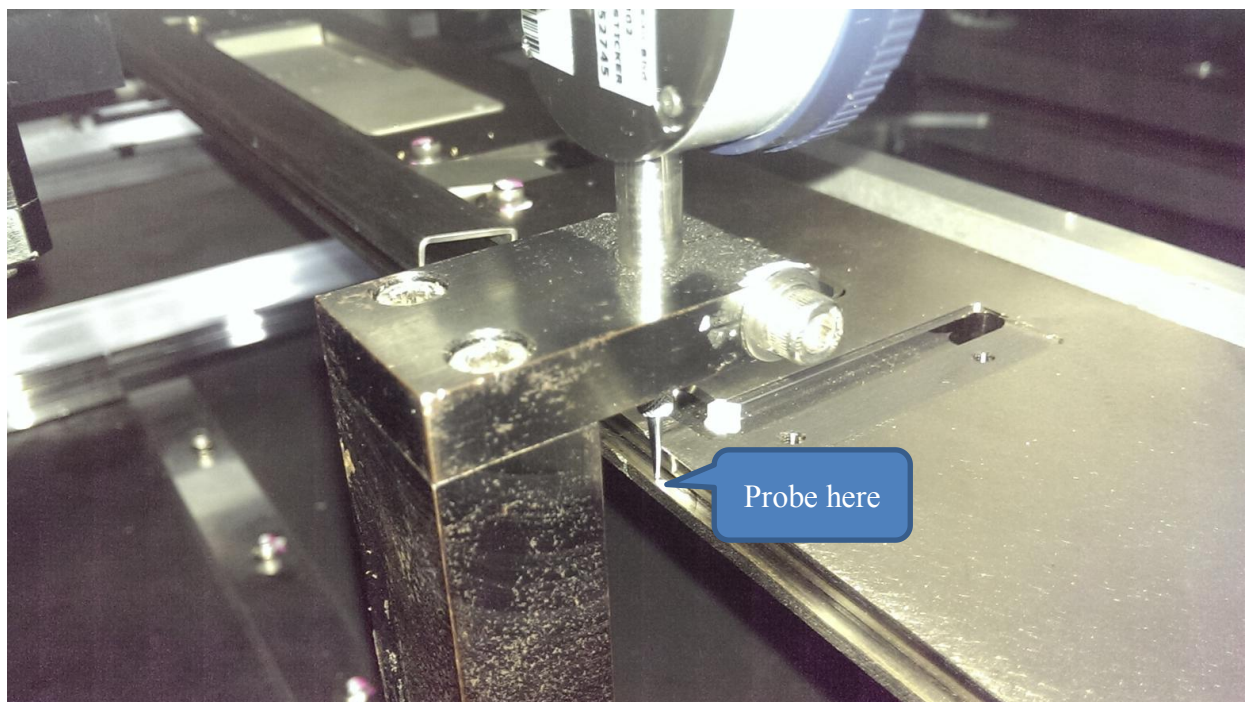


Figure 25

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28. Check the gauge reading, adjust (knocking) the rail height until the reading is same with 18. Get the height same as setting, try control the height in ± 10 micron. Gentle remind, the more needle point press, the gauge meter will move to right side. Once get the accuracy, tighten all the screws.

Knock the rail **down**, the gauge indicator will move to **left**.

Knock the rail **up**, the gauge indicator will move to **right**.

Advise tighten screw 1 by 1 and check the variation of meter, control the height until all screws is tighten.

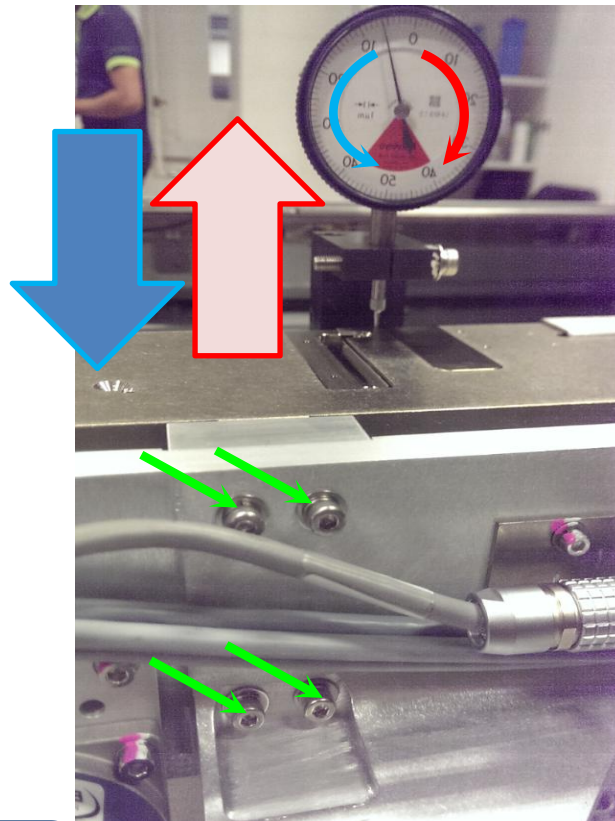


Figure 26

Current result, adjust rail height until obtain the rail height need and all screws tighten

Control the final result in ± 10 micron

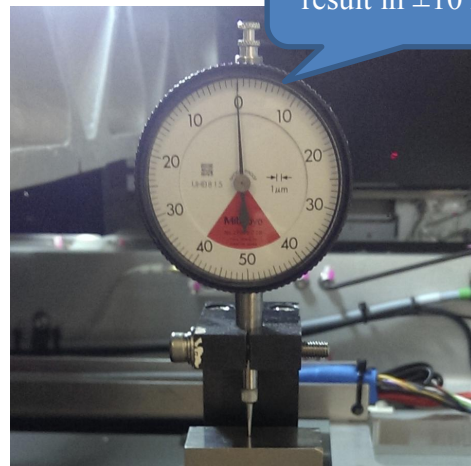


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29. Place the 4XAHW-A107 near Left PIP as shown.

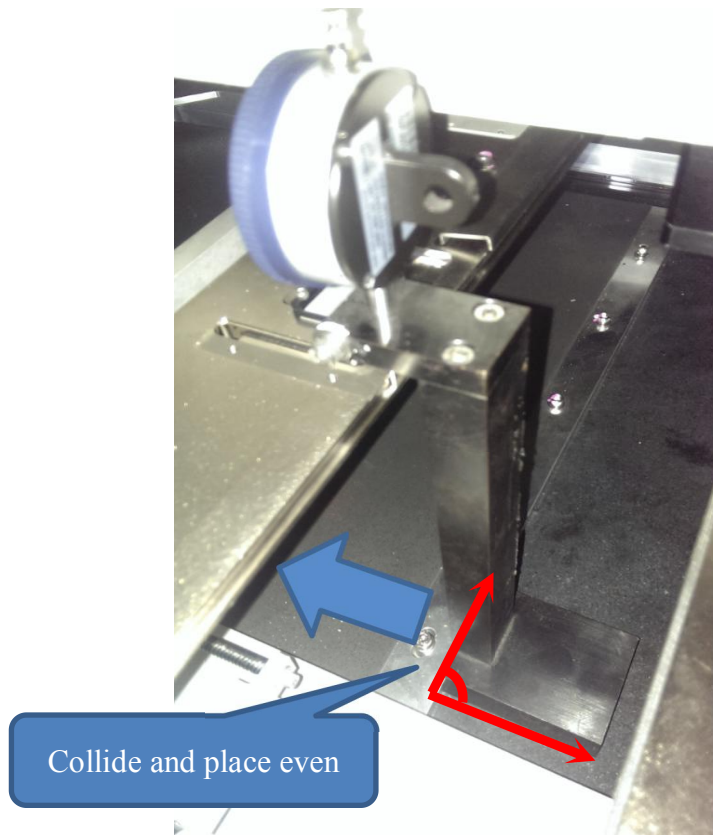


Figure 28

30. Manually move the stage until the needle point probe at inner rail surface in front on Left PIP opening as shown.

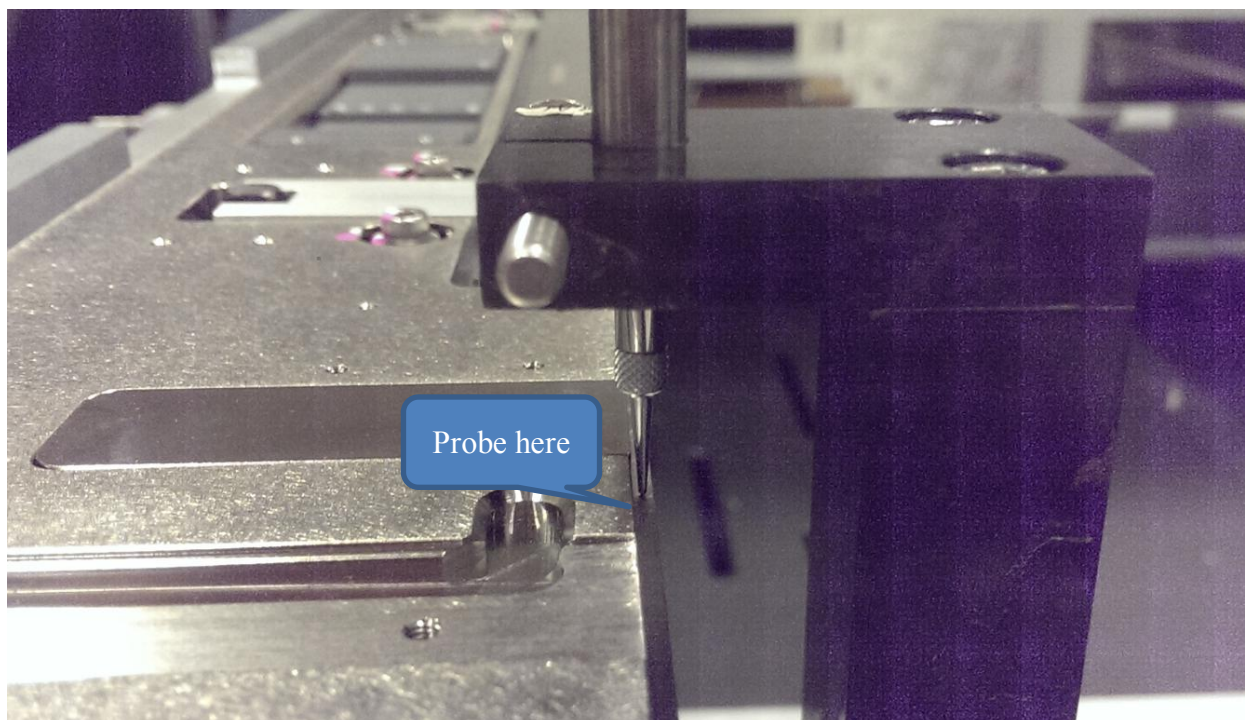


Figure 29

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31. Check the gauge reading, adjust (knocking) the rail height until the reading is same with 18. Get the height same as setting, try control the height in ± 10 micron. Gentle remind, the more needle point press, the gauge meter will move to right side. Once get the accuracy, tighten all the screws.

Knock the rail **down**, the gauge indicator will move to **left**.

Knock the rail **up**, the gauge indicator will move to **right**.

Advise tighten screw 1 by 1 and check the variation of meter, control the height until all screws is tighten.

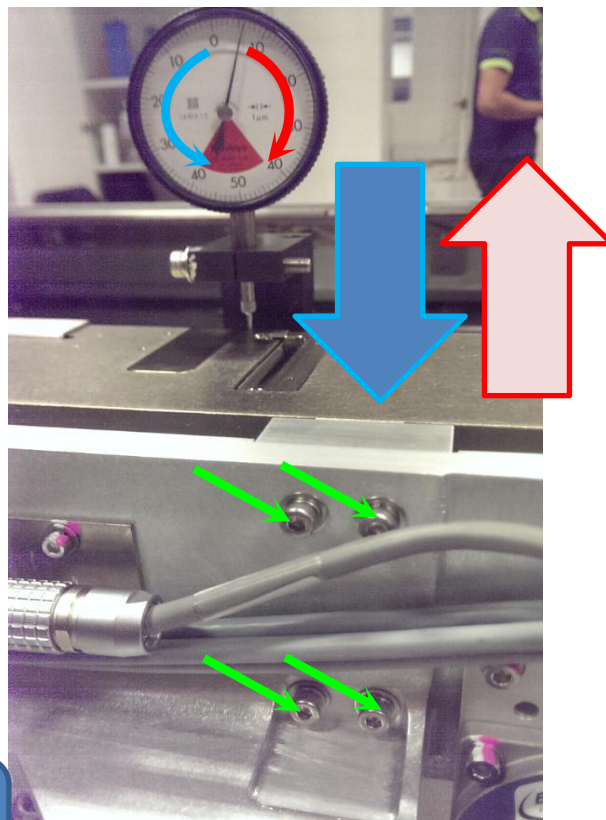


Figure 30

Current result, adjust rail height until obtain the rail height need and all screws tighten

Control the final result in ± 10 micron

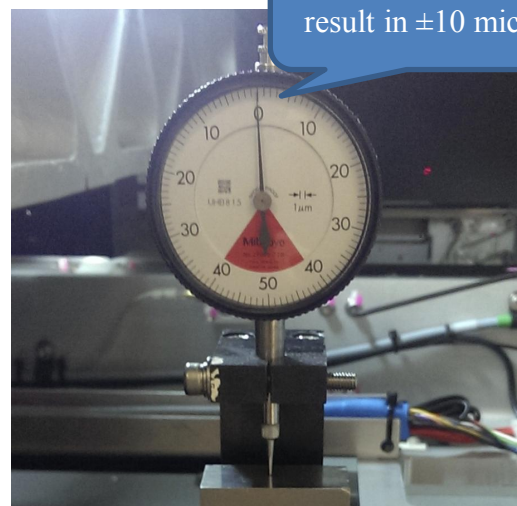
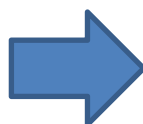


Figure 31

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32. Repeat step 25 and 26 at right side and check if the reading remain same, if out, repeat step 25 ~ 30 until both side rail height same height with 4XAHW-106 with control difference ± 10 micron. Perfectly exactly same height with 4XAHW-106.

Remark, if can't get the height as expected after, might need to re-position 4XAXY-009 x 2 depend on the situation, then repeat all step 21 ~30 for calibration.

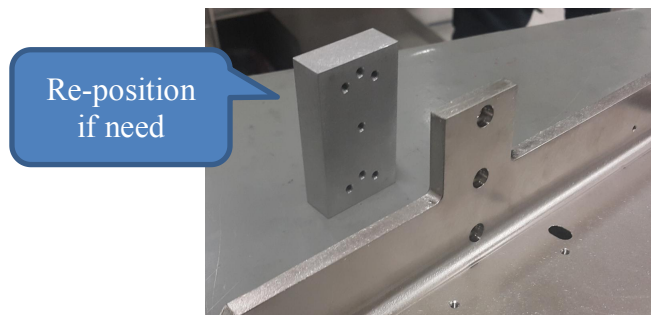


Figure 32

33. Once done, **ensure all screw at both side is fully tighten and remove all the tools and jig.**
34. Connect the machine origin tubing back to CYLINDER, 1/2 (P/N, 2810-0001) with origin position (P6 at top, P2 at bottom), **lock** the Bearing_Oiless 1/8ID X .252OD (P/N, **2650-0108**)

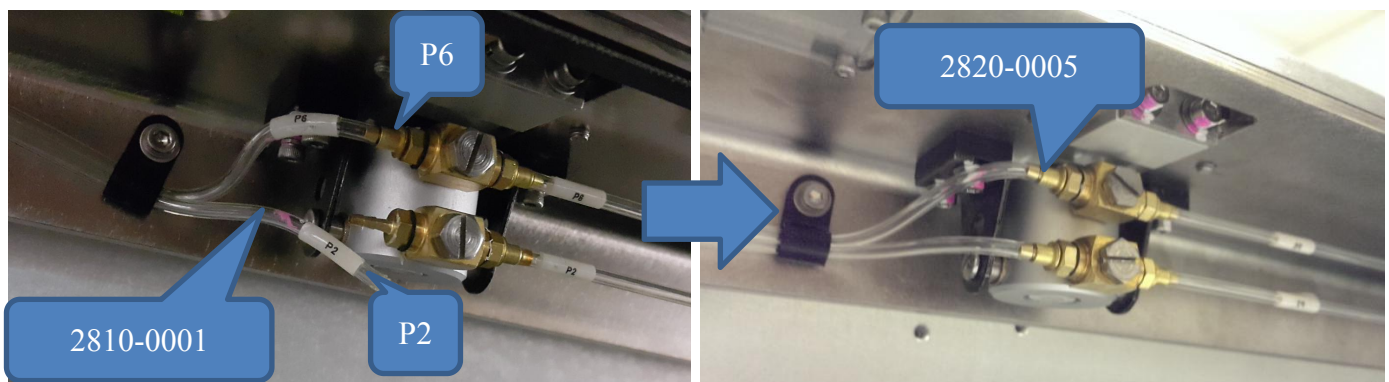


Figure 33

35. Route all the cables and air hose nicely and insert cable to 2560-0001 and screw. Tidy up with cable tie as shown.

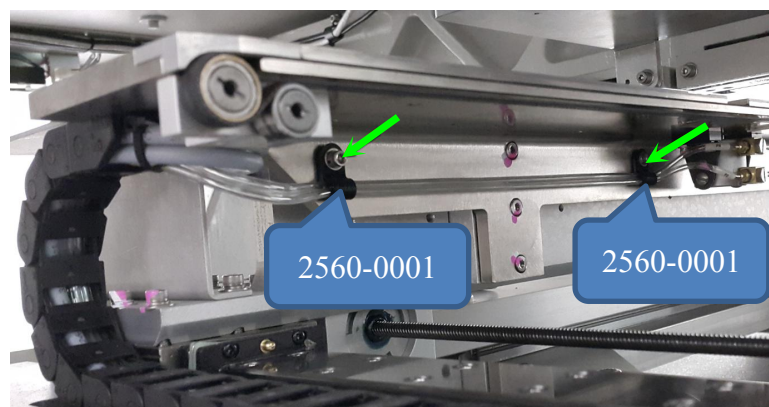


Figure 35

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36. Connect 2000-0001 back to new rail kit very 1st joint as shown, route those cable nicely.

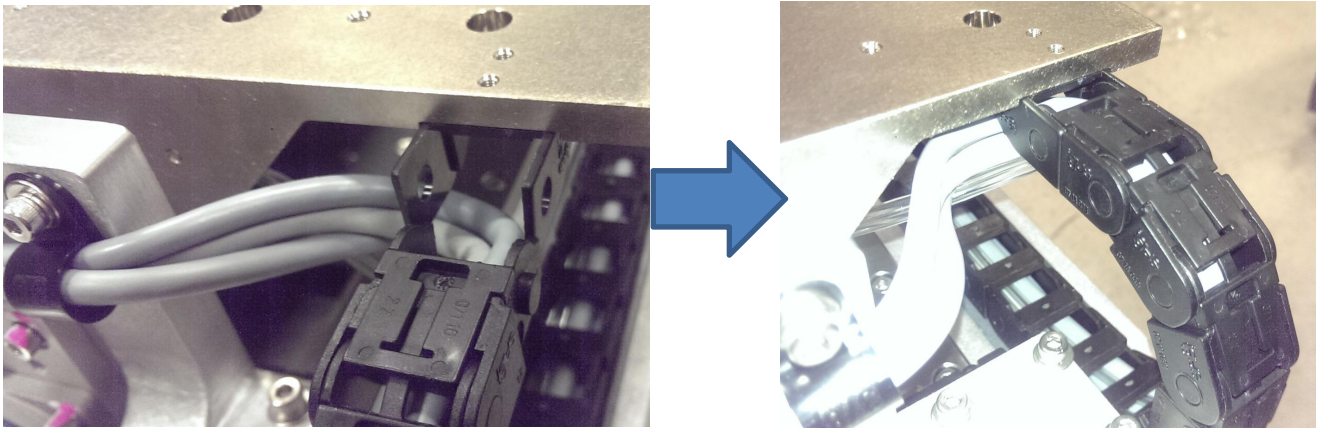


Figure 36

37. Connect the DC motor to cable.

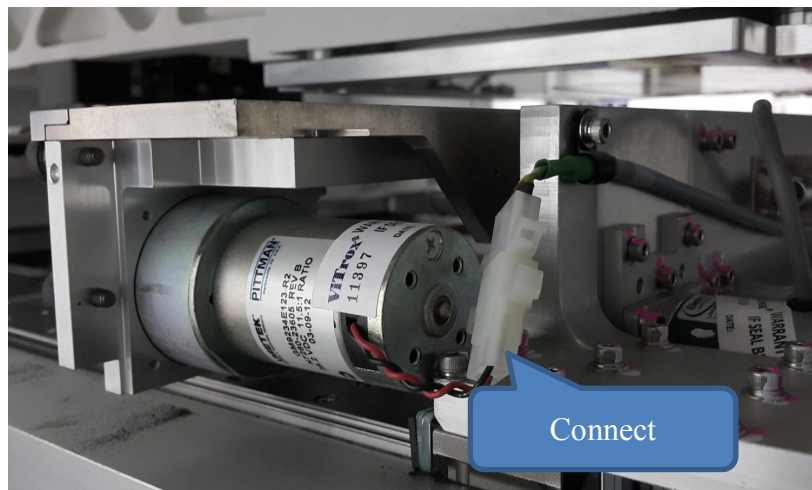


Figure 37

38. Assembly the Shield,Board Drop (P/N, 4XAGR-031) back to origin position and ensure the height is even with Field stop assembly.



Figure 38

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39. **Transfer the conveyor belt from old rail kit to new rail kit**, ensure the belt is well align to avoid burr issue. May refer to “**Belt Alignment Procedure**” for optimization if needed.
40. Ensure all tools and jig completely remove from machine inner side.
41. Assemble left outer barrier back to origin position, and tighten all the temper proof screws. Ensure the interlock switch is well align. Connect all cables back to origin and route cables nicely, ensure cable is not clamp when outer barrier open. **Cautions, door assembly is extreme heavy, need 2 people to remove.**

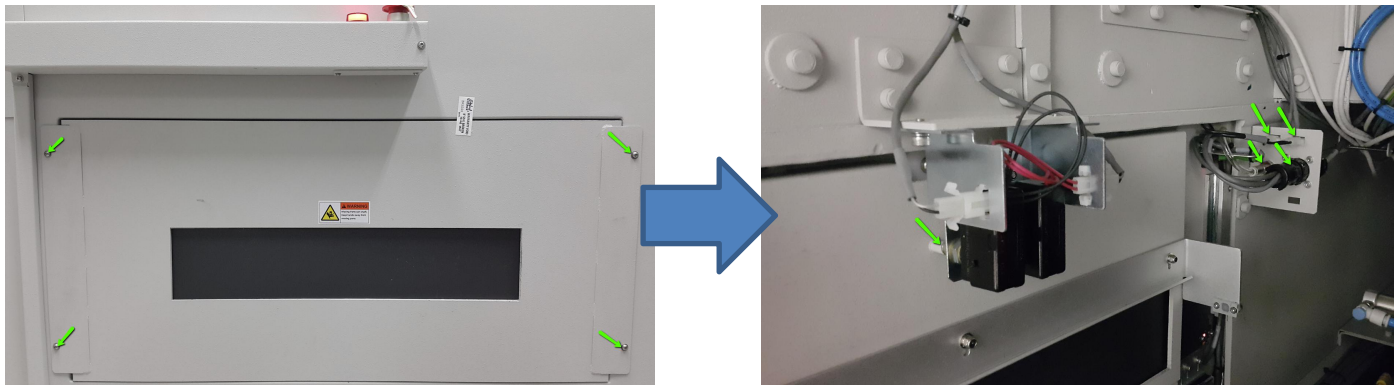


Figure 39

42. Assemble right outer barrier back to origin position, and tighten all the temper proof screws. Ensure the interlock switch is well align. Connect all cables back to origin and route cables nicely, ensure cable is not clamp when outer barrier open. **Cautions, door assembly is extreme heavy, need 2 people to remove.**

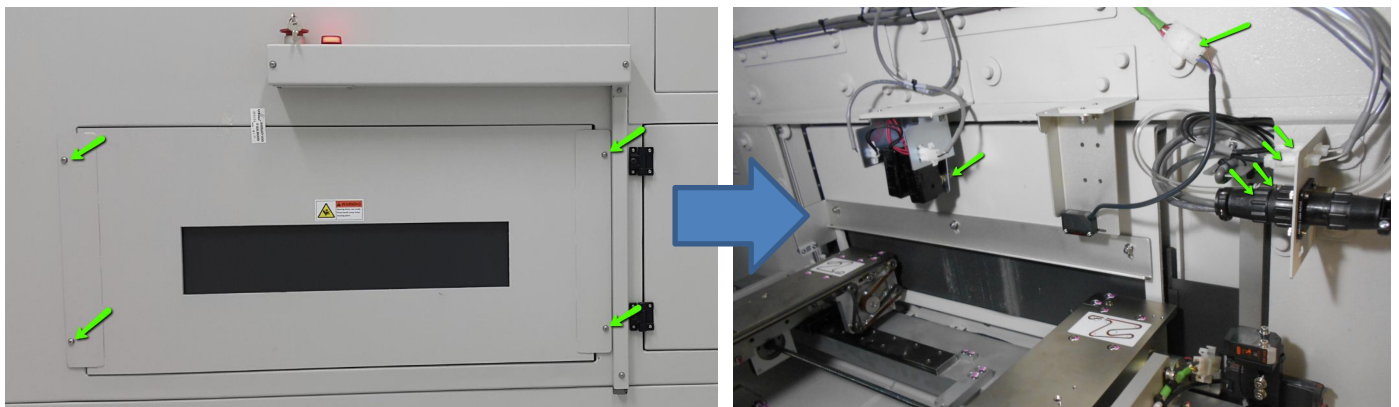


Figure 40

43. Power on machine and open the main air inlet. Start-up as usual, log in window and launch V810 GUI.

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44. Go to DIO tab and initialize. Open the front access door and manually exercise the panel clamp, **ensure command and physical move is sync, if not, swap the air hose orientation for step 34**. Check belt spinning and PIP mechanical move condition. Ensure input, output and physical move are synced.

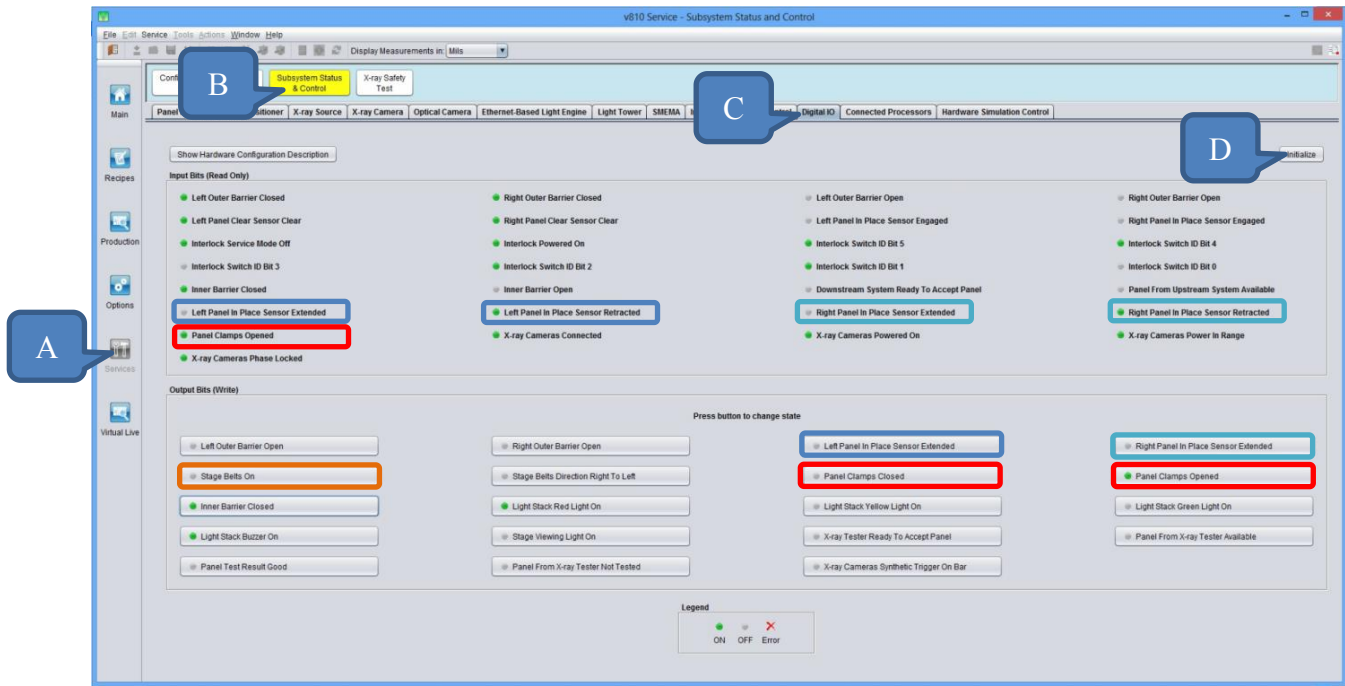


Figure 41

45. Close all access doors and go to “Panel Handling” tab, select measurement unit as “MM” and initialize, then manually set rail width to **150mm**.

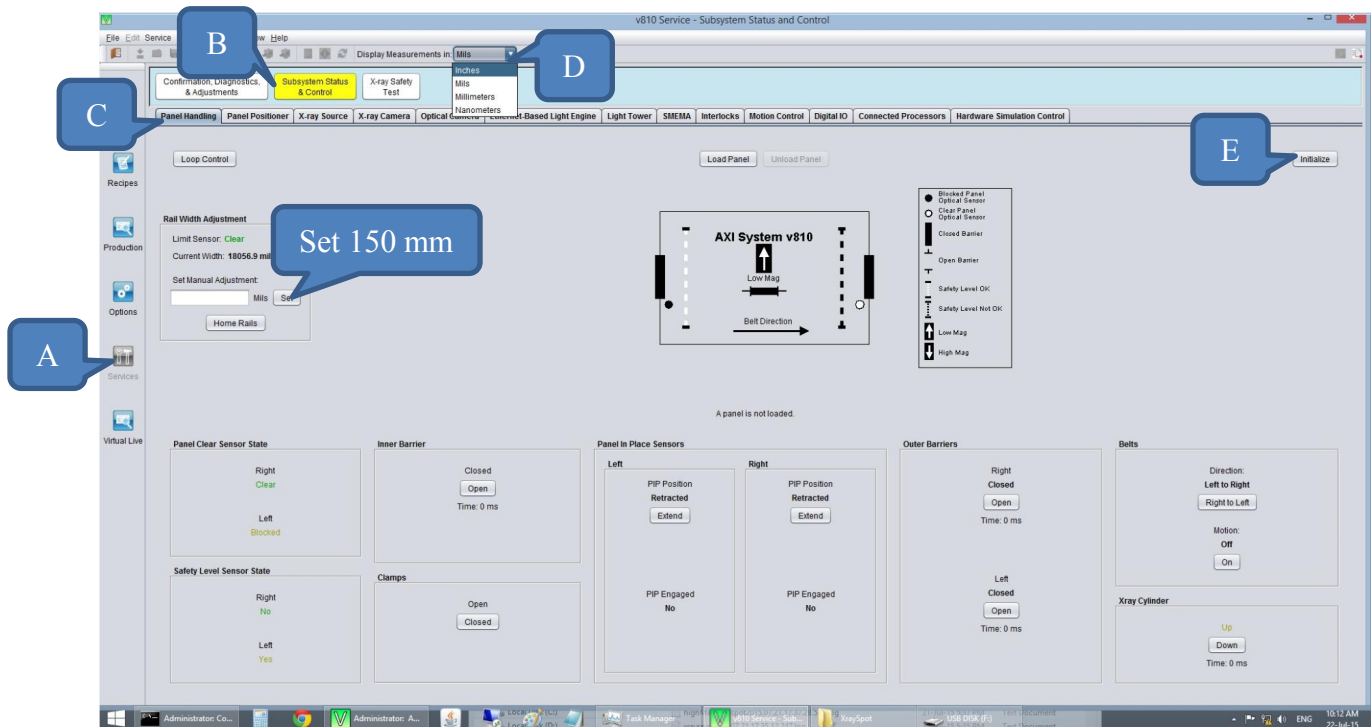


Figure 42

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46. Turn off X-ray and open front access door.
47. Measure the physical rail width with caliper near Left and Right PIP as shown. The parallelism spec should be only ± 0.125 mm. If failed, please refer to “AWA Parallelism Adjustment Procedure” for optimization.



Figure 43

48. Close all access doors and go to “Panel Handling” tab, select measurement unit as “MM” and initialize, then manually set rail width to **150mm**.

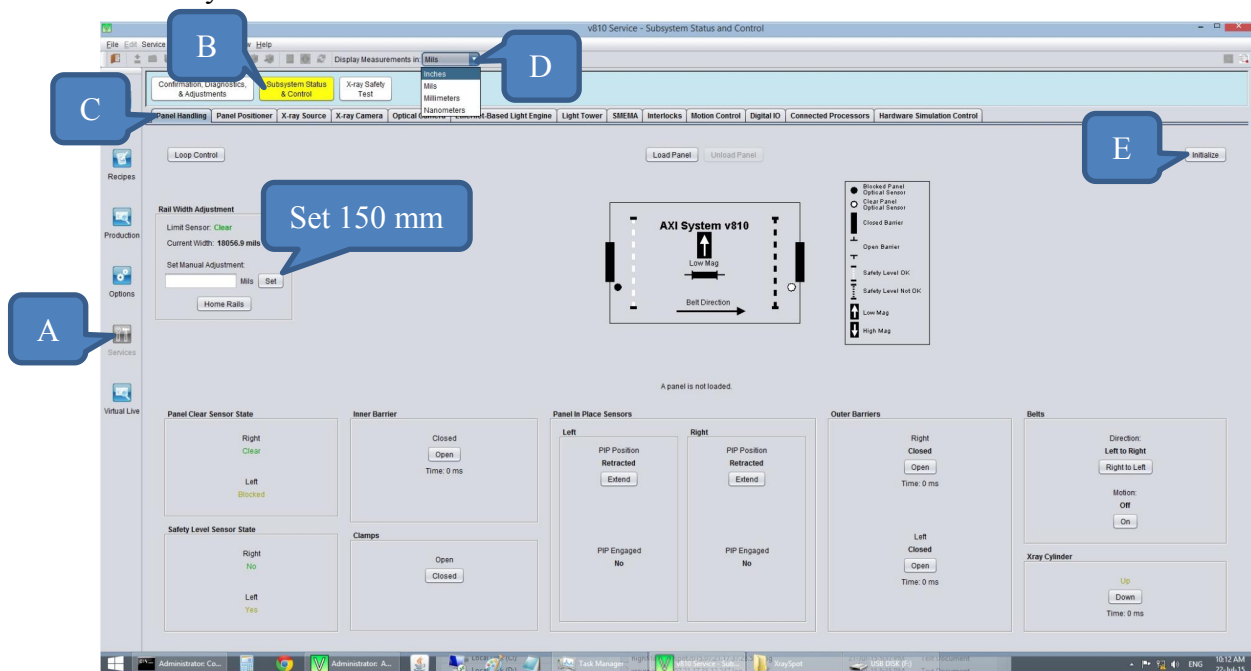


Figure 44

49. Measure the physical rail width with caliper near Left and Right PIP as shown. Physical rail width should be exactly 150mm.



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50. If step 49, physical and command not sync, open hardware.config, search for key “railWidthAxisMaximumPositionLimitInNanometers”, back-up the origin key with copy paste and add # key at beginning for the back-up key, update the difference to the origin key.
- 1mm = 1000000 nanometer
 - if key value +, physical rail width become small
 - if key value -, physical rail width become bigger

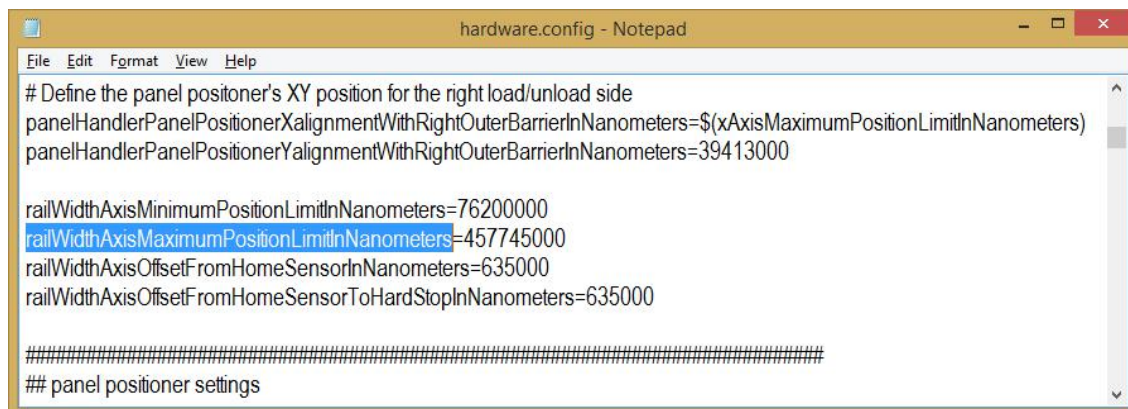


Figure 46

54. Close and restart V810 GUI to update config. Repeat step 48~49 for verification.
55. Run CDnA for 10 loops and ensure all task is passed.
- X-Ray spot is same as golden template. May refer to procedure “AXI V810 STD X-Ray Tube Alignment Procedure” for optimization.

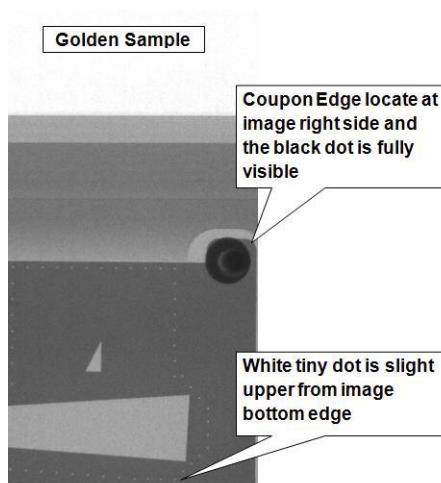


Figure 47

- Magnification spec, low mag, 5.25 and high mag 9.09. May refer to procedure “AXI V810 STD VM (19um and 11um)Tube Height Calibration” .
56. Run production and ensure no ghost image issue and collect all related log files and send to Vitrox HQ for verification.